Opportunities for Research in Air Transportation

Kenneth Button

Abstract

This article is concerned with the likely future direction of social science, and especially economic, research in air transportation. To avoid excessive speculation, the assessment is based on as much factual information as possible. It assumes, for example, that much of the technology likely to be used in the near future is known today, although it may not yet be fully developed. To provide a structure, other than just a long list of research possibilities, the study considers three basic trends—continuing trends, trend breaks, and new trends—in air transportation. The approach adopted is also mainly demand driven in that it looks at the probable research needed to explore these various trends, and it says little about how this research may be resourced, or how research priorities are likely to be determined. It is positive in its approach, trying to be as objective as possible, and focusing more on the use of developments in economic and related analysis than on normative debates about what policies should be pursued.

Keywords

air transportation, research, economics, airlines, airports, air traffic control

> Not everything that can be counted does not necessarily count;
> everything that counts cannot necessarily be counted.
>
> —William Bruce Cameron

For a variety of reasons, research into air transportation has grown significantly over recent years.[1] The number of outlets for research findings, such as academic and professional journals, conferences, and workshops for the discussion and dissemination of findings, have expanded at a

Kenneth Button
George Mason University
Kbutton@gmu.edu

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commensurable rate. The necessary material allowing for quantitative study has also grown, as have the methodology and techniques analyzing it. For example, there has been the emergence of "big data" coupled with growth in computer power and developments in programming that allows the manipulation of this data.

In addition, on the demand side, the range of challenges confronting those active in the air transportation sector and researchers outside has diversified with, for example, growing concerns over the interactions between aviation activities and the environment and, more immediately at the time of writing, the role that air transportation plays in spreading diseases. At the more micro level, appreciation of the role that air transportation can play in the development of local economies and specific industries such as tourism has expanded as the world economy has globalized. New technologies, including automated aircraft (drones), and innovative land-use planning concepts, such as Aerotropoli (Kasarda and Lindsay 2011), have added to the demands for enhanced analysis of the social implications of aviation.

This article is concerned with discussing some of the likely future research needs of the aviation community in changing conditions, and the opportunities for conducting relevant studies. Opportunities in this context will inevitably be related to the availability of suitable theory, data, and appropriate empirical estimation methods when quantification is required, or their counterparts when the needs are of a more qualitative nature. While there is some consideration in this article of emergent trends in air transportation, such as those involving urban air mobility (UAM) (Straubinger, Verhoef, and de Groot 2021) and zero carbon emission aircraft (Klöwer et al. 2021), most of the discussion of research opportunities does not go outside of the broad technical boundaries of what we now understand by air transportation; it is not a discussion of abstract "Futures."

There are always opportunities for conducting research into air transportation matters. In practice, however, there are challenges in making use of these opportunities and in getting useful general theoretical findings. Much of the research is practical, empirical, and often very contextual. Although this is sometimes seen as being second-tier to abstract theorizing, it can be of immense importance. The incredible upsurge of work from 2019 on the spread of viruses by air travelers was, for example, stimulated by the COVID-19 pandemic and not some abstract notion that the research was "needed" (see Sun et al. 2021). Much the same can be said for the use of big data to enhance load factors of aircraft and, associate with this, the

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greater fuel efficiency of aircraft and their use. This is at a time when global warming has become a key policy concern.

Our objective is neither to catalog ongoing initiatives and study areas, nor to offer a list of future research projects (and possible funding sources), but rather to focus on likely themes of public and private interest regarding future air transportation research and the approaches that promise a shedding of more light on the workings of the sector. In this context, the article is oriented toward the social sciences, and especially economics and its related subareas. There are inevitable cross-comments regarding technological challenges and developments, but these are not dealt with in any detail. To set the piece in its wider context, we begin with some discussion of potential important trends in air transportation.

Trends in Air Transportation

An oft-used structuring for a discussion of future research opportunities is to think in temporal terms. This could involve, for example, the market, short-run and long-run concepts used in economics. That is not, however, the approach favored here. Rather we look at the probable continuing trends in air transportation, breaks in these trends, and new trends. From this we consider probable research requirements as we move forward.[2] The approach clearly contains temporal elements of the more traditional approach but also appreciates that some research is fundamental and likely to be on-going, while some represents changes in direction or new paths that arise in response to current events. These latter may be either endogenous developments within the sector, say a sea change in aviation technology, or exogenous factors, such as the growth in long-distance teleworking. The trichotomy is, though, still admittedly rather fuzzy but nevertheless seems useful in our context.

Under our categorization, continuing trends refer to topics that have often been extensively researched in the past but for a variety of reasons will continue to attract the attention of researchers. They are not trends in the sense of extrapolation of, say, data series, but rather consistencies in relationships between variables. In economics this may entail trends in the relationship between air fares or income, and airline traffic—price and income demand elasticities—but can also embody persistent market imperfections, including, externalities, the underlying nature of supply and demand, and links between air transportation and economic development. Study of these long-term trends is important, not just for appreciating the underlying temporal qualitative and quantitative factors affecting

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aviation, but more widely in the determination of counterfactuals when making comparisons across air transportation networks, and when assessing the implications of trend breaks and new trends.

Examining trend breaks in air transportation research involves the study of important changes in ways air transportation is viewed and supplied. It looks at widespread deviations away from the long-term trend but within the same broad technical and institutional constraints. This has included in the past, for example, subjects such as the role of airline regulation after the enactment of the US Airline Deregulation Act of 1978, examination of aviation security after the events in the United States of September 11, 2001, and the implications of the recent COVID pandemic on the performance of the airline sector. Trend-break studies therefore consider what are essentially kinks in previously continuous trends. Put in economic terms, there is no shift in airlines' production functions but rather a change in their form. The empirical work frequently involves natural experiments and the examination of the implications of unexpected shocks on the aviation sector. Almost by definition, trend breaks are generally sudden, often affecting particular air transportation markets but leaving others largely unaffected.

New trends are frequently associated with significant improvements in technologies, the adoption of radically different business models, or shifts in the social preference function. They generally take some time to emerge and impact on air transportation. The resultant issues are difficult to foresee, although they often provide considerable subject matter for those engaged in "Futures Studies"; the systematic study of possible, probable, and preferable futures. Many new trends are associated with technology developments, such as the appearance of jet engines in aviation or GPS in location and tracking applications, or changes in political priorities. Here, we take a very conservative view and consider developments in aviation that are currently being actively researched with their technologies at least in the experimental stage. Examples may include pilotless commercial aircraft, the introduction of satellite navigation systems, advanced air mobility (AAM), and carbon-free aircraft, as well as more macro ideas in geographical planning such as the fostering of Aerotropoli. Because of the inevitable lack of hard, future data, research in this area almost inevitably involves some use of stated-preference methods, often entailing expert opinion analysis and a focus on qualitative projections.

There are no hard divisions between the various trends. The categorizations are used as a way of structuring, refining, and commenting on a

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rather wide-ranging list of research opportunities. Overlapping is inevitable, and the topics included are in no way comprehensive. One may consider this article, therefore, more of a scoping study than one making firm predictions or advocating support for any specific area of research. But before moving to these issues it seems helpful to offer a brief retrospective that sketches a very selective picture of how we have arrived at the current position regarding air transportation research.

Approaches to Air Transportation Research

When looking at the future, rather than immediately leaping in, it is often useful to review what has gone before. There have been many developments in the underlying approaches to economic and related social science studies of air transportation over the years. One development that seems especially germane at the time of writing is the considerable increase in the importance of "numbers" in analysis.

While much of the pioneering analysis of air transportation was largely qualitative in its nature, frequently with a legal or historical leaning, and often focused on case studies, more recent research has followed the general trends in social science and taken on decidedly more quantitative methodologies. The availability of more and better data (e.g., by definition time-series data get longer with time), enhanced computer power and, accompanying this, software, more sophisticated modeling, and statistical analysis have been some of the factors favoring this. But, in addition, contributing to this quantification of research has been the demands from decision-makers, both from the public and private sector sides of aviation, seeking more complete information regarding the consequences of their actions. It has also been part of a larger trend. Since the 1990s the focus of economics, and social sciences more generally, has moved away from pure theorizing to greater quantification and evaluation. To put it simply, there are far more equations and numbers in reports and articles today than thirty years or so. The advent of the computer and electronic data storage is part of the explanation, but there are other factors.

Historically, economic and related social science research into air transportation has increasingly proved a very powerful instrument, not just in intellectualizing the issues involved in the workings of the air transportation sector, but also in providing guidance and techniques for policymakers with responsibilities for evaluating outcomes. This has been especially so in areas requiring the development of analytical tools. These tools may include cost-benefit analysis for infrastructure investment appraisal of

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airports and air traffic control, data envelopment analysis and stochastic frontier analysis to assess the efficiency of transportation infrastructure (Scotti et al. 2012; Adler and Berechman 2001), as well as the effects of regulation on airline costs (Caves, Christensen, and Tretheway 1984), and stated-preference techniques to evaluate aviation safety and environmental measures.

Whether this focus on quantification as the core element in much of air transportation research will continue is not so certain. The approach has been increasingly criticized in the social sciences generally, and in economics in particular. One reason is that, in transportation in general, it has not provided the accuracy of forecasts of polices outcomes that was anticipated by many, and this includes aviation forecasts (Solvoll, Mathisen, and Welde 2020). While part of this is due to technical and data problems, people like Flyvbjerg, Skamris, and Buhl (2006) have questioned the general selectivity and interpretation of transportation forecasts, and especially those used for airport investment assessments. In particular, they question the objectivity with which forecasts are chosen. One reason for this is an increasing focus on the wider role of the institution in shaping air transportation rather than just on physical investments and market outcomes, and this can be more difficult to quantify.

Further, the majority of, largely quantitative economic analysis of air transportation has relied heavily on neoclassical economic theories with their focus on the behavior of *homo economicus*—agents who are consistently rational, narrowly self-interested, and pursue their subjectively defined ends optimally. However, as has been seen in the awarding of the Nobel Prize (the Sveriges Riksbank Prize) in economics to the likes of Richard Thaler, Daniel Kahneman, Robert Schiller, Vernon Smith, and, albeit earlier, Herbert A. Simon, the underlying assumptions of this approach are not going unchallenged as a foundation for economic analysis. Basically, human beings do not act rationally all of the time for a variety of reasons; nor do they always have full information.[3]

Homo economicus is essentially an engineering concept; people are treated as machines. Although not original to him, the methodology with its numerous analytical advantages was large adopted from Alfred Marshall's (1890) economic constructs. Marshall, however, realized its limitations in explaining what are largely biological phenomena (Thomas 1991). More recently, behavioral economists have begun injecting more realism into their analysis of human behavior, incorporating the effects of psychological, cognitive, emotional, cultural, and social factors on the

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decisions of individuals and institutions (Thaler 2016). For example, people do not have full information but act within boundaries (Simon 1978); not all managers seek full production efficiency because of inert areas (Leibenstein 1979); small "nudges" can be of consequence in decision making (Thaler and Sunstein 2008); and people have excessive confidence in human judgment (Kahnemn 2011).[4]

The relevance of this shift away from a mechanical approach to put more emphasis on human behavior in air transportation research is twofold. First, moving forward, it offers a more thorough method for examining air transportation activities. It is not incumbered by excessively restrictive prior assumptions of how agents respond to stimuli. Second, it offers the possibility of mining previous works from a different perspective. It provides the basis for reassessing their conclusions and the implications of these conclusions. This affords better hindsight into the consequences of previous policies. While in some respects behavioral approaches are best suited to qualitative analysis, this is less so than it used to be as new data and information collection methods have evolved with methodologies such as experimental economics providing a rich vein of research opportunities.[5]

Continuing Trends

Looking broadly at the pattern of commercial air transportation since the 1920s, the path is fairly clear: upward with a few periods of turbulence. But this has been accompanied by many technical and institutional changes stimulating research into the roles of such things as improved aircraft technology, more sophisticated air traffic control, changes in the regulatory environment, and greater demand as passengers' incomes have risen. Isolating the importance of each of these and other factors has been a major component of research on air transportation. It will also inevitably offer many more opportunities in the future. It is basically about the ways that markets clear in the short term and the outcomes that occur when factors exogenous to air transportation are held constant. There seem to be two broad areas that offer additional opportunities to deepen and extend our understanding of some of these continuing trends.

Cliometrics

Cliometrics, a retrospective approach to research, essentially quantitative economics history, was developed by Fogel (1964) in the context of trends in nineteenth-century US railroads. It makes use of modern quantitative methods to explore historical data. A certain amount of this type of work

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is already done in air transportation but it tends to relate to more recent periods, and thus is often not really capturing continuity in relationships. Commercial airlines, and associated activities such as air traffic control, have, however, been around since the 1920s and, just as Fogel managed to tease out some interesting results for railroads some seventy years or so after the event, so there are opportunities for economic histories to apply his methods to aviation.

The approach would appear particularly useful when exploring longer-term macro trends, for example, links between air transportation supply and economic development. There are ongoing debates, for example, over causality when it comes to investment in aviation infrastructure and growth in GDP. The opportunities for studying these issues and untangling effects have increased as statistical methods have become more sophisticated and data sources have improved and are likely to do so in the future.

In addition, cliometrics has latent potential for enhancing understanding of the air transportation markets and institutions that have to date been underresearched, and most notably those involving developing countries (Button, Martini, and Scotti 2015). As air transportation develops in lower-income nations, it is important not only to have benchmarks against which progress may be assessed, but also to have clear ideas as to the importance economic and other aviation policy initiatives have in the development process.

Replication and Meta-Analysis

The factors influencing the behavior of the various elements of air transportation – airlines, airports and air traffic control—have been well, if not comprehensively, studied. There are, however, difficulties in evaluating the quality of the empirical analysis. One would hope that an appropriate model should predict well or at least meet the needs of those who seek to use it for predictive or forecasting purposes. It should reflect the continuing trends and highlight any diversions from them. To test for this in the natural sciences, replication is an important criterion used. Basically, studies should be consistent in what they find and to support this. Studies of essentially the same phenomenon should yield statistically close approximations to each other if a relationship is robust. They should replicate.[6] There have, however, been arguments dating back at least to the 1980s that more replication is needed in the social sciences (Kane 1984) and these apply to aviation analysis.

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As in much of social science research, replication studies of results from prior aviation analyses are very limited. A number of factors mitigate against conducting the necessary work. Independent, direct replications of others' findings can be time-consuming and take resources away from other projects that are considered as more original, and perhaps of greater importance. The original data and, in the case of statistical work, the coding used are not always available to compare contemporary findings with previous analysis. The prior studies examined may also have used a diversity of control variables and of model specifications. On the demand side, replications are harder to publish, often because they are not seen by the academic community as very novel. In other words, there is something of a void in the scientific assessment of previous work on air transportation issues.

One possible quasi-replication method of comparing empirical work looking at the same issue is to use meta-data analysis. This entails statistical synthesizing of the results of prior quantitative studies adjusting for their context (Stanley and Jarrell 2005). While potentially useful, care is necessary when applying meta-analysis to air transportation networks which can take a variety of forms and involve diverse geographical spreads (Button 2019a). Meta-analysis is also not "a scientific technique." Subjective decisions are required regarding which studies to include and, in particular, to allow for variations in the quality of individual pieces of research. Meta-data analysis has limitations in synthesizing quantitative works based on stated-preference methodologies because each prior study tends to be based on unique "data"-gathering techniques. Simply put, meta-data analysis may add pseudo-scientific exactitude to assessments if not used carefully.

Having said this, a number of meta-analyses have been conducted regarding air transportation matters, but mainly on more specific issues than those under consideration here. These include Nelson's (2004) and Schipper, Nijkamp, and Rietveld's (1998) syntheses of on airport noise nuisance issues, Li, Harris, and Chen's (2007) work on airline safety, and Gallet and Doucouliagos's (2014) on the income elasticity of air travel.

Much of this meta work, and indeed other syntheses, related to air transportation has relied primarily on examining the academic literature. While there are crude methods of assessing the quality of this work, for example by looking at citation counts, these do not offer completely satisfactory ways of weighting the outputs of these previous studies. Certainly, there are few meta-analyses that have delved back into the raw data when selecting prior works for inclusion.

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There are also input omissions. In particular, gray literature (e.g., reports and working papers of which there are many in air transportation) and publications not written in English tend to be omitted. There is inherent bias in what is written. There is, for example, a tendency for editors of peer-reviewed journals to favor results that support existing theories or contain findings broadly conforming to "expected" magnitudes and to reject as outliers papers where findings radically differ from the norm. In the social sciences the lack of any template for reporting results offers an added challenge, as does the focus on quantitative findings, with little attention on the qualitative. Despite these limitations, the mining of prior research efforts seeking out common factors that affect the air transportation variable of interest seems to provide a useful way for assessing the factors influencing continuing trends in aviation. There are good reasons for more research in this area.

Trend Breaks

Natural Experiments

Natural experiments can offer opportunities for researching underlying relationships in air transportation markets and for assessing the implications of policy actions. Such experiments are empirical studies in which individuals or groups are exposed to experimental and other conditions that are outside the control of the investigators.[7] The events ("exposures") of interest are observational and are not *controlled* as with *randomized experiments* such as with stated preference studies. Natural experiments are most useful when there has been a clearly defined exposure, say to a new regulation that comes into force that affects a distinct sub-population but leaves the rest of the population unaffected. Any resultant change in the behavior of affected population may then plausibly be attributed to its exposure.[8] Resultant changes in outcomes in the two subpopulations may plausibly be attributed to their exposure or not. In this sense, the difference between a natural experiment and a nonexperimental observational study is that the former includes a comparison of conditions that pave the way for causal inference, while the latter does not.

Two relatively recent opportunities that created potential bases for natural experiments in aviation have been the September 11, 2001, attacks in the United States and the 2020 COVID pandemic.[9] Both events were outside of the control of researchers, and both led to significant breaks to trends in some aviation markets or submarkets while other markets

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were hardly affected. Neither has been exposed to extensive and rigorous quantitative analysis of the extent these events impacted the trending path of air transportation, although it is admittedly at the time of writing rather early to expect assessment of any COVID effect. Techniques such as differences-in-differences analysis now provide a statistical methodology for quantifying the implications of such events on trend patterns.

Institutional-based examples of natural experiments in air transport often involve regulatory reforms. Almost since its inception, virtually all aspects of aviation have been heavily regulated. Much of this regulation has been seen as being in the "public interest" for reasons of safety, market stability, environmental protection, restraints on market power, national protectionism, and so on. In many cases economic regulations were justified on grounds that they reduce wider transactions costs (Spiller 2013). Regulations, including such things as subsidies, are in this case designed to ensure a stable supply of air services that would stimulate private-sector investment in a specific geographical area or in a particular economic sector such as tourism. Countries often change their regulatory structures allowing researchers opportunities to compare their experiences of economic instability with others that have retained the status quo.

Whether regulation actually served these purposes has, however, been questioned. Along the lines of Stigler's (1971) and Peltzman's (1976), neoclassical economic arguments have been used to maintain that market interventions have been in the interests of those whose actions are regulated and/or those with governance responsibilities regarding the enforcement and interpretation of regulations. In other words, they have not served the "public interest."

Economic and business research has played an important role in clarifying these situations. Academics including Keeler (1972), Levine (1965), and Jordan (1970) offered ex-ante evidence, for example, that the regulation of interstate US domestic airline pricing and market entry prior to the 1978 Airline Deregulation Act was producing high fares and was severely limiting capacity. Outcomes that were more beneficial to airlines than the general public. From another direction came research seeking to define what a deregulated airline market was likely to look like in the long term—for example, Baumol's (1982) work on contestability. Despite these ex-ante analyses, the exact outcomes of the 1978 reforms, while generally in the direction anticipated, threw up some unexpected results (Kahn 1988).

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Business Models and Industrial Structures

Airline business models are continually changing as new markets emerge and old ones contract, as new technologies are developed, and as managerial ideas change. From the perspective of economics, as opposed to business administration, these changes are studied as part of micro-micro economics that looks "inside of the firm" (Leibenstein 1979).[10] The number of case studies focusing on airlines' internal workings, the ways that small groups of carriers coordinate their behavior within alliances, and, more recently, joint ventures is relatively small. Individual company and business histories can pinpoint the factors leading to success of carriers or to their downfall.

One issue involving the nature of firms and their dynamic influences on markets entails notions of "disruptive innovation." This does not just mean that successful market entry is disruptive, it generally is, but rather the disruption caused is of a specific form. Bower and Christensen (1995), when defining the term, and reinforced in Christensen, Raynor, and McDonald (2015), have been very clear about what disruptive innovation involves, "In our experience, too many people who speak of 'disruption' have not read a serious book or article on the subject." This is because they see disruptive competition as involving entry by a small company into a market where established incumbents were focusing on providing products for their most demanding and profitable customers. A disruptive entrant targets neglected customers by offering more suitable functionality and usually lower prices. Response from incumbents is initially lax because they are focusing on different markets, and this allows the newcomer to move up the market taking mainstream business from the incumbents.

What constitutes a disruptive innovation in practice, however, is not that simple. For example, the entry of Southwest Airlines in the US national market and of Ryanair in European scheduled airline markets may be seen as acting in this way. They met the demands of passengers seeking low-fare basic products that incumbent-legacy carriers were not serving. Further, they subsequently developed their business model allowing encroachment on markets served by the incumbents—elements of disruption outlined by Christensen and his coauthors. But most airline markets prior to LCC entry were far from free markets with regulations inhibiting many actions of the incumbent airlines. The types of controls placed on carriers—including route licensing, lack of ability to price discriminate, and rate-of-return fare setting—were largely the reasons they behaved the way they did rather than their actions being purely market driven.[11]

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New Trends

Inevitably the arrival of any aviation-related innovation, be it institutional or technical, has some degree of overlap with other forms of mobility. It affects their continuous trends. In some cases, the effect may be extreme, for example, leading to the destruction of the market for an existing mode; the advent of the motor car caused the demise of the horse-drawn carriage being an example. In other cases, they may complement existing modes leading to a shift up in existing trends; the advent of wide-bodied aircraft and the growth in long-distance travel and tourism being an example.

Big Data

The tools of research will be different in the near future when contrasted to those of the past but only to a limited extent. "Big data," or data analytics (Button 2019b), for example, is playing an increasingly important role in the managing of air transportation (Dou and Tan 2020; Abdulsamie, and Ramadan Al-Azab 2021). Airlines enhance use of their capacity by tailoring the service-fare packages they offer closer to the need of individual customers, and structure their scheduling of planes, crew, and maintenance to reduce costs.

Big data involves ways of analyzing and systematically extracting information from or otherwise dealing with data sets that are too large or complex to be dealt with using traditional data-processing methods. For example, big data is used by airlines, through machine learning, to evaluate passenger demand across routes networks, to gain insights to optimize aircraft ground handling, and to improve yield management when fare setting. It can also, by analyzing customers' preferences, enhance the effectiveness of airline loyalty (frequent flyer) programs, as well as leading to improved routing of planes (Kasturi et al. 2016), and the tracking of both baggage and freight. It is used in air navigation systems for weather forecasting and routing, by aircraft maintenance companies for near real-time monitoring of the performance of planes in service. As an example of the latter, Boeing analyzes over 2 million conditions daily across 4,000 aircraft providing billions of observations as a part of its Airplane Health Management system. Airports use big data to assess passengers satisfaction with their services (Papagiannopoulos and Lopez 2018) and to evaluate capacity utilization (Wan et al. 2020).

Because of its relative newness, however, big data is of limited use in looking at continuing trends. It is currently mainly an input into real-time analysis in business applications and for cross-sectional work, but as big

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data bases inevitably expand with time, they have considerable scope for looking at the dynamics of aviation markets.

From an economic perspective, however, questions arise concerning the ways in which data are gathered and subsequently used. For example, is it exploiting consumers who find themselves at an informational disadvantage to the air transportation service providers (Mandlik and Kadirov 2018)? It also may add to an already considerable bias toward that which can be counted, to hark back to the opening quote, and a continued discounting of softer information and data.

Economic Regulation and Air Transportation Innovation

Regulation governing air transportation may have two opposite effects. It may stymie future development by pushing up the costs of providing transportation services or it may foster faster growth by stimulating cost-reducing innovations (Aghion, Bergeaud, and Van Reenen 2021). Regarding the former, for instance, Storrs Hall (2021) credits the product liability-costs introduced by the US authorities in the 1970s with killing the manufacture of private planes. Before 1978 the US federal regulatory structure fostered an inefficient domestic airline industry that retrospectively can be seen to have held back both the creation of hub-and-spoke networks and the development of low-cost carriers.

Much of the hardware used in air transportation (the "metal") is, almost by definition, highly mobile, and this mobility extends across geographical boundaries. To allow efficient interoperability, common standards can be important. This also extends to human capital including airline crew and air traffic controllers where individuals interact with common technologies. Standardization, however, can act as a dampener on subsequent technical progress if it results in substantial stranded costs for those adopting it. In other words, the transfer to the new trend can be difficult and involve large transaction costs if it is from a standard base.[12]

Currently, the challenges of adapting to satellite-based air navigation systems as replacements for established radar-based systems—for example, NextGen in the United States (Strohmeier et al. 2014) and the Single European Sky in the European Union (Lawless 2014)—reflect some of the problem transitioning to a new trend. Previous research, based mainly on comparative statics, provided limited guidance as to the economic gains from the new system, and especially did little to explore the transition costs themselves. We still remain largely unsure of what additional innovations will ensue once NextGen and its likes are fully implemented.

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Urban Air Mobility

Air transportation has traditionally been an interurban mode despite efforts to develop urban machines. Alberto Santos-Dumont's airship No.6 was, for instance, the first gasoline-powered dirigible balloon, and he won the Deutsch Prize in 1901 by flying around the Eiffel Tower on it. It had all of the basics required for urban air transportation. It was a vertical takeoff and landing machine (VTOL); it was propelled through the air, had horizontal steering and could vertically climb or descend in flight, but the technologies used were not advanced enough for the dirigible to be a viable urban mode.

While helicopter services have provided a degree of air mobility above large cities such as New York and São Paulo, these carry a minute share of urban traffic and are costly as well as being environmental intrusive. Advances in electric motors and batteries, remote-control systems, and airframes have seen a renewed interest in urban air mobility (UAM) (Straubinger et al. 2020; Cohen and Shaheen 2021). Drones, for example, are coming into use for providing local delivery services, and they are seriously being considered for adoption as more sophisticated airport shuttles and air taxis (Goyal et al. 2021).

Much of the regulatory framework for governing the design and use of UAM vehicles derives from that relating to conventional aircraft. But this seems to be a stop-gap measure. From the perspective of public administration, there will inevitably be a need for separate certification processes for unmanned aircraft as well as regulation of infrastructure supply and use. The persons involved in flying the equipment, either actually or virtually, will need qualifications. The pricing and property rights over air space have to be defined and a governance structure established for their use, including pricing and air traffic management. There are issues of equity involving mainly, but not exclusively, environmental and safety concerns. This is often in the context of where flights occur and at what altitude. Added to this there may be welfare, distributional concerns involving environmental costs and access to the UAM services to consider to ensure that notions of environmental justice are addressed (Straubinger, Verhoef, and de Groot 2021).

Important from an operational as well as regulatory perspective, UAM may serve many purposes and take a variety of forms. Phasing in of new technologies is influenced by both market and institutional factors, such as regulations as noted above (Reiche et al. 2018). It is still unclear as to the optimal ways for doing this, and uncertain as to the obstacles that may be encountered.

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Low and Zero Carbon Technologies

The concern with the macro impact of aviation on the global environment is not strictly a new trend; it has been around since at least the appearance of the *Brundtland Report* (World Commission on Environment and Development 1987), but the general concept of sustainable development has gradually been morphing into a series of more concrete actions and proposals.

The deregulation of many airline markets, and the resultant efforts of airlines to reduce their cost in the face of competitive pressures, has initiated more efficient use of carbon fuels by airlines (Button et al. 2015). *Brundtland* added to these pressures, causing both national governments and international agencies to seek ways of bringing about greater reduction in carbon emissions. Measures have included the use of the European Union's emissions trading system (Scheelhaase et al. 2018, Scheelhaase 2019) and the International Civil Aviation Organization's adoption of carbon offset policies (Scheelhaase, and Maertens 2020).

In part these actions are a social response to a better scientific understanding of the contribution of civil aviation emissions to global climate change (Klöwer et al. 2021), but also reflect an attempt to use economic instruments to address at least some of the challenges involved. But efforts to try to understand the implications of these policies, both with regard to aviation and with regard to the users of aviation, have been limited to date.

Conclusions

Research seldom follows a regular pattern, and this has been manifestly so in the context of work regarding air transportation. The opportunities to undertake research, and the topics chosen together with the approaches used, are governed by the challenges confronting society and the resources that are available. There are also legacy effects with much of the current research conducted in air transportation being the result of past research. Tomorrow's research will almost inevitably be led by that of today. As with virtually any area for study, air transportation research generates far more questions than there are resources to examine them. Prioritization is sometimes attempted when government or big business is concerned, but the criteria selected and the ways the criteria are used are seldom consistent. Additionally, priorities are contextual depending on time and place.

This article has set the scene for areas where there will likely be opportunities for economic and related research in air transportation in the sense that there are serious voids in our knowledge. Although there are some allusions to specific projects, the focus is rather on the broad approaches to research that will be needed, given the continuing trends in the sector,

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trend breaks that are occurring, and the new trends that are emerging. Yet the future is highly uncertain, and the research needs and opportunities in air transportation will produce many surprises.

Notes

1. The terms *air transportation economics* and *aviation* are used interchangeably in the article although research involving the aircraft production industry is largely set aside. The terms also only refer to civil air transportation. The article does not cover private or military aviation although there are interfaces at several levels, including air traffic control and the regulation of drones, that may offer opportunities for research initiatives. It also, to keep the length of the piece within bounds, focuses almost exclusively on passenger transportation. This is not to say that air cargo, an industry that carriers about 40 percent of world trade by value, does not throw up an array of opportunities for research, but these are not discussed.
2. Economists may think in terms set out by institutional economist Oliver Williamson (2000). Continuing trends are the everyday activities that clear markets, the interactions of supply and demand. They continually function within established structures setting the parameters for market clearing. One can then think of trend breaks as situations whereby there is some fundamental change in these structures, perhaps institutional or perhaps technical, while new trends emerge when these structures shift, resulting in new paradigms with entirely different functions.
3. Heggie (1978) made this, as well as several other related points when discussing surface transportation analysis.
4. Nudges have yet to be used in the aviation sector, but field experiments suggest that they can be effective in such things as pilots reducing their fuel burn (Gosnell, List, and Metcalfe 2016).
5. One of the first "experiments" in economics involved the allocation of airport landing slots (Rassenti, Smith, and Bulfin 1982).
6. As with many academic ideas, defining replications is not straightforward. Clemens (2017) defines *replication studies* as those that use the same sample, population, and empirical specification as opposed to reproduction studies that use different samples from the same population but use the same specification; reanalysis uses the same sample and population but different specifications; and extensions use different samples and populations but the same specification.
7. The methodology is often traced back to Card and Krueger's (1994) work on minimum wages rates.
8. The works of Keeler, Levine, and Jordan cited earlier that compared trends in less-regulated US intrastate with more-regulated interstate scheduled air services may be seen as pioneering uses of this methodology.
9. The surge in air transportation research that followed events such as the COVID pandemic, the Great Recession of 2007–9, and the accidents involving the Boeing 737 Max aircraft give support to the Italian Renaissance philosopher Niccolo Machiavelli, when he said, "Never waste an opportunity offered by a good crisis."

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10. One might think of examples such as Amankwah-Amoah and Debrah's (2014) historical analysis of Air Afrique or, in a slightly different way, Sedláčková and Lokaj's (2017) analysis of two small airline alliances.
11. Christensen, Raynor, and McDonald (2015) also make this point regarding the emergence of Uber and other transportation network carries. They filled gaps left by highly regulated taxicab markets rather than by the market per se.
12. A nontransportation example was the adoption of the QWERTY keyboard on mechanical typewriters. This reduced the frequency of keys jamming. Later, however, as electric typewriters were developed with keyboards that allowed faster finger speeds, the layout slowed their uptake because of the high costs of retraining typists (Hossain and Morgan 2009).

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